

# DESIGN AND ANALYSIS OF ALGORITHMS

## Practical No. 6: Dijkstra's Shortest Path Algorithm

### 1. Aim

To find the shortest path from a given source vertex to all other vertices in a weighted graph using Dijkstra's algorithm.

### 2. Objective

- Understand single-source shortest path computation.
- Select the nearest unprocessed vertex using a greedy strategy.
- Determine shortest distances and predecessor paths.

### 3. Theory

Dijkstra's algorithm solves the single-source shortest path problem for graphs with non-negative edge weights. It repeatedly selects the unvisited vertex with the smallest tentative distance and relaxes its adjacent edges. With an adjacency matrix the complexity is  $O(V^2)$ ; with an adjacency list and binary heap it is commonly  $O((V + E) \log V)$ . It is not suitable for negative edge weights.

### 4. Conclusion

Dijkstra's algorithm efficiently determines shortest paths from a source when all edge weights are non-negative. The practical demonstrates greedy selection and edge relaxation.